

Twenty-five top-tier structural conjectures in number theory

Mechanisms, phase boundaries, and proof interfaces

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Abstract

This manuscript gives exactly twenty-five candidate conjectures in five programmes: connected prime fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal arithmetic dynamics, and adelic factorization processes. Every numbered statement is intended to be independently research-worthy: it introduces a canonical operator, process, invariant, classification, or phase boundary rather than merely recording a familiar Gaussian, Poisson, Gumbel, Bateman–Horn, or Poisson–Dirichlet consequence. Each entry states its mechanism, nearest literature boundary, first decisive theorem, and a failure mode that would reveal missing mathematics. The collection is screened against the cited literature but does not claim professionally certified priority or truth.

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1 Top-tier admission standard

A statement is counted among the twenty-five only when it passes all eight gates below.

- T1. Canonical object.** The ensemble, normalization, topology, and order of limits are fixed before data are inspected.
- T2. Structural mechanism.** The claim identifies an arithmetic, geometric, spectral, or adelic source rather than merely a limiting distribution.

- T3. Independent content.** It introduces an invariant or transformation not already contained in another numbered statement.
- T4. Discrimination.** At least two serious models make different predictions.
- T5. Leverage.** Truth would organize a family of problems, not one statistic.
- T6. Counterfactual value.** Falsity would expose a missing invariant, obstruction, or scale.
- T7. Proof interface.** A smaller theorem is visible and mathematically natural.
- T8. Novelty boundary.** The nearest known theorem or conjectural framework is named and the additional content is stated explicitly.

Evidence convention. The companion script is deterministic and deliberately modest. It verifies exact identities, searches for elementary counterexamples, compares local models, and reports finite-range tensions. No computation is offered as proof or priority certification.

2 Index of the twenty-five conjectures

No.	Name	Independent top-tier content
1	Connected motif generating functional	A full connected diagram calculus for all joint cumulants of prime motifs.
2	Complete overlap-renormalization filtration	A classification of every possible covariance scale and rigid eigenspace in a finite motif family.
3	Mesoscopic local-spectral trace formula	Independently defined Euler-product and multiple-zero functionals agree after common counterterms and smoothing.
4	Arithmetic polymer Gibbs law	Conditional prime-motif clusters form a canonical Gibbs polymer process after the low-zero intensity is divided out.
5	Topological expansion of non-Gaussianity	The first non-Gaussian terms are graded by the homology of overlap-incidence complexes.
6	Connected first-arrival functional	The entire least-prime point process has connected kernels obtained from same-class prime correlations.
7	Gauss-polyspectral reciprocity	Additive and multiplicative polyspectra of the least-prime field are exact finite transforms of one another and have dual arithmetic descriptions.
8	Rubinstein–Sarnak decorated terminal law	The last residue classes form a decorated Cox process directed by a canonical low-zero Gaussian field.
9	Nonlinear spectral response calculus	Low-zero and exceptional-zero perturbations pass through first-arrival statistics by universal Volterra response operators.
10	Local-information capacity of odd class groups	Cohen–Lenstra independence has a sharp two-resource boundary governed by cell entropy and conductor complexity.
11	Kummer–Haar law on relation tori	Saturated multiplicative relations give the exact horizontal support and Haar law of Fermat-quotient vectors.

No.	Name	Independent top-tier content
12	Rank-dimensional finite-logarithm large sieve	A positive power range of frequencies obeys a large sieve whose dual dimension is saturated rank.
13	Mesoscopic-to-lattice shrinking-target transition	Haar universality persists for growing targets, while bounded lattice targets acquire a separate arithmetic regulator intensity.
14	Horizontal p -adic regulator-matrix law	Unit regulator matrices are Haar on their exact relation spaces, with determinantal rare-event laws.
15	Functorial horizontal logarithms on tori	A finitely generated global subgroup of a torus has a functorial matrix-valued finite-logarithm Haar law.
16	Entropy-conductor universality of arboreal resolution	The observable depth of an open arboreal representation is asymptotic to $\log_d \log X$ and is finite-index stable.
17	Arboreal family large sieve	Growing preimage quotients satisfy a square-root Frobenius large sieve over parameters and primes.
18	Coloured dynatomic Galois-factor process	Exact-period components generate independent coloured, Frobenius-marked factor processes, with growing-degree support restrictions.
19	Primitive valuation process for dynatomic values	The full primitive valuation vector has an exact adelic product law retaining every bad-prime and collision state.
20	Divisor-sensitive dynamical gcd classification	Positive normalized gcd height is exactly eventual entry into a preperiodic subvariety carrying the relevant common divisor.
21	Frobenius-marked Poisson-Dirichlet process	Macroscopic polynomial-value factors carry fixed-point-size-biased Galois marks.
22	Three-scale polynomial factorization	Small valuations, mesoscopic scale-invariant factors, and macroscopic marked factors form one conservation-corrected product structure.
23	Adelic Gibbs gluing for reducible values	Reducible polynomial factor processes are an archimedean- p -adic Gibbs mixture of independent component processes.
24	Full multivariate adelic saddle law	Joint smoothness is governed by a complete joint action and Hessian, not a scalar Euler correction.
25	Buchstab-Bateman-Horn component flow	Rough prime, semiprime, and higher-almost-prime polynomial values follow universal Buchstab components with Galois marks.

3 Programme I: connected prime-pattern fields

Let $H \subset \mathbb{Z}$ be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \quad Y_H(x, L) = \sum_n w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where w is a fixed smooth compactly supported function and $\mathfrak{S}(H)$ is the Hardy-Littlewood singular series $\prod_p (1 - \nu_H(p)/p)(1 - 1/p)^{-|H|}$, with $\nu_H(p)$ the number of residue classes occupied by H modulo p . The window position x is sampled from $[X, 2X]$ with density $dx/(x \log 2)$ and $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. For translated motifs $H_i + t_i$, their *exact overlap type* records every equality among shifted prime constraints. The connected singular series $\mathfrak{S}^c(K_1, \dots, K_r)$ is obtained by Möbius inversion over set partitions of the indexed tuple $(K_1, \dots, K_r)$.

Conjecture 1 (Connected motif generating functional). *Fix admissible motifs $H_1, \dots, H_m$ and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp \left(\sum_{i=1}^m z_i Y_{H_i}(x, L) \right).$$

For every fixed total degree R , the degree- $\leq R$ Taylor polynomial of $\log \mathcal{Z}_{X,L}$ has a uniform asymptotic expansion indexed by connected exact- overlap diagrams of at most R translated motifs. A diagram whose coincidence hypergraph connects the r translated sets $H_{i_1} + t_1, \dots, H_{i_r} + t_r$ contributes, at the leading order of its stratum,

$$W_\tau(L; \mathbf{t}) \mathfrak{S} \left(\bigcup_{j=1}^r (H_{i_j} + t_j) \right) \prod_v (\log X)^{m_v-1},$$

where $W_\tau(L; \mathbf{t}) = L \int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L) du$ is the archimedean window-overlap weight of the diagram and m_v is the multiplicity of the prime constraint at v : any set partition that separates a coincidence loses at least one factor of $\log X$, so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to $\mathfrak{S}^c(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$ with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of L and $\log X$, the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.

Why this is top-tier. This is a multitype connected field theory for prime constellations. It simultaneously organizes covariance, odd moments, mixed motifs, overlap singularities, and lower-order prime-power effects. Montgomery–Soundararajan and Kuperberg provide the closest one-field moment frameworks, while constrained singular-series sums provide the nearest local input [1, 3, 4]. The new content is the complete connected multitype functional. The first decisive theorem is the full third-cumulant formula for two distinct pair motifs.

Conjecture 2 (Complete overlap-renormalization filtration). *Fix a finite motif family $H_1, \dots, H_m$. Apply the degree-two specialization of Conjecture 1 and order all of its connected covariance strata by their asymptotic scales $s_1 \succ s_2 \succ \dots$ (lexicographically in powers of L , $\log X$, and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of L and of logarithms differ only by powers of $\theta = \log L / \log X$ and merge into a single stratum whose limiting form is polynomial in θ . The filtration, the graded convergence, and the eigenvalue asymptotics below are asserted for every fixed $\theta \in (\varepsilon, 1 - \varepsilon)$, with each $A_\nu = A_\nu(\theta)$ nondegenerate for every such θ . Recursively subtract the earlier strata and let A_ν be the limiting covariance form at scale s_ν , and put*

$$\mathcal{W}_\nu = \bigcap_{\mu < \nu} \ker A_\mu.$$

For every fixed truncation of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$, and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to s_ν times a positive eigenvalue of the induced form A_ν . Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 1.

Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 1, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.

Why this is top-tier. The claim classifies all mechanisms by which a finite linear combination of prime motifs can become rigid. It predicts a canonical renormalization filtration, not merely one variance formula, and gives an algorithm for constructing observables that expose deeper arithmetic strata. A counterexample would identify a genuinely new source of rigidity outside overlap and disjoint correlation. The first theorem is a three-motif example in which one overlap form has a rational kernel and the next form is nondegenerate on it.

Conjecture 3 (Mesoscopic local–spectral trace formula). *Fix motifs H_i, H_j and a Schwartz function ϕ with $\phi(0) = 0$. Let $Q_{ij}^{\text{loc}}(\phi; L)$ be the exact-overlap-renormalized singular-series form*

$$\sum_{t \in \mathbb{Z}} \left[L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t).$$

Independently, let $\Lambda_T^\sharp$ be the symmetrically truncated Weil explicit-formula distribution: its pairing with every compactly supported smooth test function is the standard pole term minus the sum over nontrivial zeros with $|\Im \rho| \leq T$, together with the trivial-zero and archimedean terms, while the omitted prime-power contribution is retained as its named remainder. Define $Q_{ij}^{\text{spec}}(\phi; X, L; T)$ by replacing every von Mangoldt factor in the smoothed motif covariance by $\Lambda_T^\sharp$, expanding the product, and applying the same exact-overlap Möbius projection as on the local side. Thus the two forms are defined independently, with pole, trivial-zero, diagonal, and prime-power terms carried explicitly.

The translation sum is finite, since the compact support of w makes the window overlap vanish for $|t| \gg L$. For every sufficiently large fixed A , take $T = (X/L)(\log X)^A$. After adding the explicit tail of the spectral truncation,

$$Q_{ij}^{\text{spec}}(\phi; X, L; T) - Q_{ij}^{\text{loc}}(\phi; L) = o(V_{ij}(X, L))$$

uniformly for $X^\varepsilon \leq L \leq X^{1-\varepsilon}$ and for ϕ in bounded Schwartz sets, where V_{ij} is the first nonzero disjoint covariance scale. The identity holds jointly for every fixed matrix of motifs and test functions and is unchanged if A is increased.

Why this is top-tier. This is a genuine trace formula: the Euler-product and multiple-zero sides are defined independently and compared only after common smoothing and named counterterms. It extends the Goldston–Montgomery and Chan prime/zero equivalence from one prime-counting field to products of shifted von Mangoldt functions [24, 2]. Failure would reveal a mesoscopic covariance functional missing from both Hardy–Littlewood and known spectral correlation languages. The first theorem is the pair-motif covariance with one factor unshifted.

Conjecture 4 (Arithmetic polymer Gibbs law for prime motifs). *Fix an admissible motif H . A finite set $A \subset \mathbb{Z}$ of occurrence starts is connected when the graph on A with an edge between u and v whenever $(H + u) \cap (H + v) \neq \emptyset$ is connected, and a polymer is a finite connected set A such that $U_A = \bigcup_{t \in A} (H + t)$ is admissible. In the truncated multiple-zero expansion used in Conjecture 3, let $Z_{X,L,R}$ be the vector of all zero monomials contributing to the one-point motif intensity whose total spectral height is at most RX/L , and let $\lambda_{X,L,R}(t)$ be the one-point intensity functional it determines at the start t . The pair of $Z_{X,L,R}$ and the occurrence process converges jointly in law as $X \rightarrow \infty$, and all conditioning below is on the limiting mode vector, a nondegenerate random object, not on a sigma-field generated at finite X , where conditioning on deterministic functions of the window position would degenerate to the sample point.*

Write $e(A) = |A| |H| - |U_A|$ for the number of shared prime constraints of a polymer, so $e(A) \geq |A| - 1$ for connected A by the multiplicity rule of Conjecture 1. For every fixed polymer A , under the limiting joint law,

$$\frac{\rho^{(|A|)}(A \mid Z)}{(\log X)^{e(A)} \prod_{t \in A} \lambda(t)} \longrightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)^{|A|}}$$

in probability as $R \rightarrow \infty$, uniformly for bounded polymer diameter. The polymer activities therefore carry the factors $(\log X)^{-e(A)}$, every non-singleton activity vanishes in the limit, and the cluster structure is a finite- X triangular-array phenomenon rather than a nondegenerate limiting decoration. The conjectured law is the first-order extremal-index expansion: the cluster-start fraction of the occurrence process obeys

$$\theta_X(H) = 1 - \frac{c(H)}{\log X} + o\left(\frac{1}{\log X}\right), \quad c(H) = \sum_{t \neq 0} \frac{\mathfrak{S}(H \cup (H+t))}{\mathfrak{S}(H)},$$

the sum running over the shifts sharing exactly one prime constraint, with the higher-order polymer corrections carrying higher powers of $1/\log X$ through the analogous connected activities. The finite- X Gibbs polymer expansion built from these activities is projectively consistent, its cluster partition function is subcritical for every fixed admissible H , and the rescaled occurrence process is, to first order, a Cox process of Poisson starts directed by the low-zero one-point intensity with the polymer corrections entering at the scale $1/\log X$.

Why this is top-tier. The conditional normalization is what makes the law canonical: the low-zero field controls only the varying one-point intensity, while complete union singular series control the residual cluster geometry, at the explicit $(\log X)^{-e(A)}$ activity scale that the multiplicity rule forces. The law uses polymers of every size: for $H = \{0, 6\}$, starts 0, 6, 12 already form a legitimate three-occurrence polymer, and for that motif $c(H) = 2 \mathfrak{S}(\{0, 6, 12\}) / \mathfrak{S}(\{0, 6\})$. A proof would connect Hardy–Littlewood constants, explicit-formula fields, Palm theory, and extreme gaps in one object.

Conjecture 5 (Topological expansion of non-Gaussian prime cumulants). *For every connected exact-overlap diagram in Conjecture 1, form the bipartite incidence complex whose vertices are motif occurrences and distinct surviving prime constraints, with an edge for incidence. Let g be its first Betti number after forced coincidences are contracted. Once all larger coincidence strata have been subtracted, a diagram of Betti number g is smaller than a tree diagram with the same external motifs by at least $(\log L)^{-g}$. The complete leading coefficient at each logarithmic order is the sum of the connected diagram weights with the corresponding Betti number. In particular, the first nonzero odd cumulant of a balanced motif statistic, a combination Y_c whose coefficient vector annihilates every coincidence-stratum form of Conjecture 2, is carried by incidence trees. For the single-prime field the fully spread connected part $\kappa_{2m+1}^{\text{sp}}$, defined by subtracting every coincidence stratum as in Conjecture 2, obeys*

$$\kappa_{2m+1}^{\text{sp}} = (c_m + o(1)) L^m (\log L)^{m+1},$$

where c_m is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 1, which carry powers of $\log X$ and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg’s singular-series sums.

Why this is top-tier. This predicts a topological genus expansion for prime statistics: every independent cycle costs a logarithmic order. Kuperberg’s odd-moment conjecture is the one-field calibration, and the incidence-homology filtration with its motif-dependent coefficients is the new mechanism [3]. A failure would show that logarithmic suppression is controlled by an arithmetic invariant not visible in diagram topology.

4 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus q and $a \in \mathbb{F}_q^\times$, let $p(q, a)$ be the least prime in the class a and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q-1}, \quad \mathcal{N}_{q,a} = \sum_{p \equiv a \pmod{q}} \delta_{\text{Li}(p)/(q-1)}.$$

The probability space chooses q uniformly among primes in $[Q, 2Q]$ and then chooses a uniformly from $\mathbb{F}_q^\times$. Write $e(x) = e^{2\pi i x}$.

Conjecture 6 (Connected first-arrival functional). *Let $w_{r,Q}^c$ be the connected factorial-cumulant measure of the random point process $\mathcal{N}_{q,a}$. For every fixed $r \geq 2$ there is a locally finite signed measure K_r on $\mathbb{R}_+^r$ such that*

$$(\log Q)^{r-1} w_{r,Q}^c \implies K_r$$

vaguely, and K_1 denotes the limiting intensity measure, which in this clock is Lebesgue measure on $\mathbb{R}_+$. The measure K_r is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of r primes constrained to one residue class. For every nonnegative compactly supported f in a fixed uniform ball on which the Laplace functional is analytic,

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^r (1 - e^{-f(u_j)}) dw_{r,Q}^c(\mathbf{u}),$$

and the asymptotic expansion obtained by inserting the limits K_r is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy $(K_r)_{r \geq 2}$.

Why this is top-tier. This is an arithmetic point-process theory of least primes. The coupon-collector model of Li–Pratt–Shakan supplies the nearest extremal analogy, but not a connected Laplace functional [5]. Truth would determine waiting times, occupancy covariance, cover times, and terminal clustering from one hierarchy. The first theorem is the existence and explicit evaluation of K_2 on compact time intervals.

Conjecture 7 (Gauss-polyspectral reciprocity for the least-prime field). *Set $T_q(0) = \text{Li}(q)/(q-1)$ and centre the resulting field f_q on all of $\mathbb{F}_q$. Let*

$$A_q(\xi) = q^{-1/2} \sum_{a \bmod q} f_q(a) e(-a\xi/q), \quad M_q(\chi) = (q-1)^{-1/2} \sum_{a \neq 0} f_q(a) \overline{\chi(a)}.$$

On the subspace of fields vanishing at 0 with zero mean the Gauss transform is an explicit unitary matrix, and the recorded value at 0 enters through an explicit rank-one correction, and together these give an explicit Gauss matrix U_q , invertible between the centred subspaces on which the two transforms are faithful, with $\mathbf{M}_q = U_q \mathbf{A}_q$, and the tensor identity below needs only linearity. Hence, for every $r \geq 2$, the full cumulant tensors obey the exact finite identity

$$\text{Cum}_r(\mathbf{M}_q) = U_q^{\otimes r} \text{Cum}_r(\mathbf{A}_q).$$

The usual additive constraint $\sum \xi_j = 0$ and multiplicative constraint $\prod \chi_j = 1$ are contractions of these full tensors. The constrained slices alone are not asserted to determine one another.

After dyadic averaging and normalization by the first nonzero connected scale, the full additive cumulant tensors converge to the Fourier transforms of the connected arrival kernels K_r of Conjecture 6 carried on the zero-sum frequency sublattice, that is to $\delta_{\sum_j \xi_j \equiv 0} \otimes \widehat{K}_r$. Applying the exact Gauss tensors before passing to the limit gives the same limiting contractions as the r -fold explicit-formula correlations of Dirichlet L -function zeros on the multiplicative side. Thus the local arrival kernels and character-zero correlations are two exactly compatible tomographic representations of one non-Gaussian field.

Why this is top-tier. The $r = 1$ Gauss identity is elementary. The conjecture is the complete higher-order reciprocity between arithmetic first-arrival cumulants and character spectra. It supplies a finite-field analogue of a trace formula and makes every mixed higher moment computable from either side. A failure would identify information destroyed by all additive and multiplicative polyspectra despite their exact tensor duality.

Conjecture 8 (Rubinstein–Sarnak decorated terminal law). *Assume GRH and a q -aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet L -functions. Let x_q satisfy $\text{Li}(x_q)/(q-1) = \log(q-1)$ and, for $A > 0$, define the centred low-zero field*

$$G_{q,A}(a) = -\frac{1}{\sigma_{q,A}} \Re \sum_{\chi \neq \chi_0} \overline{\chi(a)} \sum_{\substack{L(1/2+i\gamma, \chi)=0 \\ |\gamma| \leq (\log q)^A}} \frac{x_q^{i\gamma}}{1/2 + i\gamma},$$

where $\sigma_{q,A}^2$ is the average over reduced classes of its variance under the random-phase law, the same convention as in the compensator below. Under the random-phase law, its finite-dimensional distributions converge, first as a random prime modulus $q \in [Q, 2Q]$ tends to infinity and then as $A \rightarrow \infty$, to the Rubinstein–Sarnak Gaussian field at terminal scale.

Let b_Q be determined by $\mathbb{E}_q \# \{a : T_q(a) > b_Q\} = 1$ and define the random directing measure

$$\Lambda_{q,A}(dg \, dx) = \frac{\theta_{\text{occ}} e^{-x}}{q-1} \sum_{a \in \mathbb{F}_q^\times} e^{G_{q,A}(a) - \frac{1}{2} \text{Var}(G_{q,A}(a))} \delta_{G_{q,A}(a)}(dg) \, dx.$$

Here $\theta_{\text{occ}} = \mathbb{E}^{\text{Palm}}[|\mathcal{A}|^{-1}]$, with $\mathcal{A}$ the terminal polymer cluster, is the cluster-start fraction of the terminal polymer law, the analogue of $\theta(H)$ in Conjecture 4. The pair of $G_{q,A}$ and the point process $\sum_a \delta_{(G_{q,A}(a), T_q(a) - b_Q)}$ converges jointly in law along the same successive limits, the directing measures $\Lambda_{q,A}$ converging, under the random-phase hypothesis, to a nondegenerate log-Gaussian directing measure Λ_G . Conditionally on the limiting field, the cluster-start process is Poisson directed by Λ_G , and each start carries, at finite modulus, the terminal analogue of the polymer corrections of Conjecture 4, entering at the scale $1/\log q$. No conditioning on a sigma-field generated at finite q is invoked, since the field is a deterministic function of the sample modulus and such conditioning would degenerate. Thus the cover time, final class profiles, and their prime-race marks are projections of one decorated Cox process.

Why this is top-tier. Rubinstein–Sarnak theory describes prime-race distributions for fixed moduli [6]. This conjecture places the terminal least-prime problem in a q -aspect Gaussian environment and predicts the complete decorated extreme process, not a scalar Gumbel shift. It sharply distinguishes iid occupancy, scalar mixing, and a field-valued spectral environment.

Conjecture 9 (Nonlinear spectral response calculus for arithmetic first arrivals). *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a, t) = \sum_{\chi \in \mathcal{C}_Q} v_{Q,\chi}(t) \chi(a), \quad \|V_Q\|_T \rightarrow 0,$$

where $\mathcal{C}_Q$ has uniformly bounded cardinality and, on every fixed interval $[0, T]$, the canonical norm is $\|V\|_T = \sup_{a, 0 \leq t \leq T} |V(a, t)|$. Assume that the connected hierarchy of Conjecture 6 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed $\|\cdot\|_T$ -ball. If $S_Q(a, t)$ and $S_Q^{(0)}(t)$ are the perturbed and ordinary survival functions, then for every fixed R

$$\log \frac{S_Q(a, t)}{S_Q^{(0)}(t)} = \sum_{m=1}^R \mathcal{R}_m[V_Q^{\otimes m}](a, t) + o(\|V_Q\|_T^R)$$

uniformly on compact t -ranges. The causal Volterra operator $\mathcal{R}_m$ is the universal linked-cluster contraction of the full hierarchy $(K_r)_{r \geq 1}$ with m marked perturbation insertions, and truncating the connected hierarchy at order J gives a convergent approximation as $J \rightarrow \infty$.

If the input character support is $\mathcal{C}_Q$, the m th response is supported exactly on products of at most m input characters (subject only to coincidences and the trivial character). In particular the linear response preserves character rank, while the first new harmonics occur at quadratic order with coefficients determined by connected three-point arrival correlations.

Why this is top-tier. This is a nonlinear transfer calculus from zeros to first-hit statistics. A Siegel zero, a cluster of low zeros, and an ordinary prime-race bias become different inputs to the same response operators. The exceptional-zero rank-one statement is a special case, while the second-order harmonics provide a direct falsification test. The first theorem is the Fréchet derivative $\mathcal{R}_1$ from the two-point occupancy kernel.

Conjecture 10 (Local-information capacity of odd class groups). *Fix an odd prime ℓ . Sample negative fundamental discriminants $D \in [-2Y, -Y]$, conditioned on a fixed coarse two-primary invariant. For a set of odd primes S_Y , put $M_Y = \prod_{p \in S_Y} p$, let $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$ be the attainable sign vector, and let H_Y be the logarithm of the number of attainable cells. Let $\delta_Y(\sigma)$ be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed two-primary data, and let $\mathcal{D}_Y$ be the ambient discriminant family. Write*

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y| \delta_Y(\sigma)} - 1 \right|$$

over attainable cells with $|\mathcal{D}_Y| \delta_Y(\sigma) \geq \log Y$.

There is a sequence $\varepsilon_Y \rightarrow 0$, independent of S_Y in the subcritical range, such that uniformly over those cells

$$d_{\text{TV}}(\mathcal{L}(\text{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, 2\text{-data}), \mathcal{L}_{\text{CL}, \ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every $\varepsilon > 0$, the conclusion is uniform whenever

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq (\tfrac{1}{2} - \varepsilon) \log Y, \quad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most $\log 3 \leq \log p$ of cell entropy, $H_Y \leq \log M_Y$ throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At $H_Y \geq (1 + \varepsilon) \log Y$ no uniform statement is possible. Up to the conductor scale at which Δ_Y becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

Why this is top-tier. Wood’s framework treats fixed local conditions [7]. The new statement proposes the full information capacity of an odd class group under a growing local sigma-algebra, with the entropy governing impossibility and the conductor governing equidistribution. It would transfer directly to Selmer groups, ray class groups, and unramified extension statistics. The first theorem is uniformity for $|S_Y| \rightarrow \infty$ with $M_Y = Y^{o(1)}$.

5 Programme III: finite logarithms, regulators, and algebraic tori

For $p \nmid a$, write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple $g = (g_1, \dots, g_s)$, let $R(g)$ be its multiplicative relation lattice and let $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$ be the identity component of its annihilator. Its dimension is the saturated multiplicative rank.

Conjecture 11 (Kummer–Haar law on relation tori). *Let $\Gamma \leq \mathbb{Q}^\times$ be finitely generated, saturated, and torsion-free, and let g be any ordered generating tuple. As p varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

equidistribute with Haar measure on $\mathbb{T}_g$. More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of $\mathbb{T}_g$. The law is invariant under changing generators and depends on Γ only through its saturation and the fixed Kummer conditioning data.

Why this is top-tier. Katz formulated horizontal Wierferich equidistribution principles for algebraic groups, and Shparlinski proved estimates in different horizontal and averaged regimes [8, 9]. The new content here is the exact saturated relation torus, full generator invariance, and disintegration under fixed Kummer classes. It turns multiplicative rank into a geometric support theorem rather than a heuristic count of coordinates.

Conjecture 12 (Rank-dimensional large sieve for finite logarithms). *Choose a Smith-normal-form basis of the character lattice $X^*(\mathbb{T}_g) = \mathbb{Z}^s/R(g)^{\text{sat}}$. There is a constant $\delta_\Gamma > 0$ such that, for every $\varepsilon > 0$, every $M \leq X^{\delta_\Gamma}$, and all complex weights a_p supported on $X < p \leq 2X$,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^{\text{rank } \Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent δ_Γ is invariant under changing the generating tuple.

Why this is top-tier. This is the quantitative engine behind Conjecture 11. It predicts that saturated rank is not only the support dimension but the exact dual dimension in a prime-varying large sieve. A proof for any positive power range would control genuine shrinking targets and would be far beyond currently available fixed-frequency tests. Failure would reveal an unrecognized spacing or energy obstruction among finite-logarithm vectors.

Conjecture 13 (Mesoscopic-to-lattice shrinking-target transition). *Let $\Gamma \leq \mathbb{Q}^\times$ be saturated and torsion-free of rank r , and choose a basis of Γ , identifying its relation torus with $(\mathbb{R}/\mathbb{Z})^r$. For integers w_p with $1 \leq w_p = o(p)$, let $E_p(w)$ be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most w_p .*

Mesoscopic universality. If $w_p \rightarrow \infty$, then in every range in which the Haar mass

$$\lambda(X) = \sum_{p \leq X} \left(\frac{2w_p + 1}{p} \right)^r$$

tends to infinity, the count of $E_p(w)$ is asymptotic to $\lambda(X)$. On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.

Lattice transition. *For every fixed finite target set $B \subset \mathbb{Z}^r$, there is an arithmetic regulator intensity $\kappa_{\Gamma,B} \geq 0$ such that*

$$\#\{p \leq X : q_p|_{\Gamma} \in B\} = (\kappa_{\Gamma,B} + o(1)) \sum_{p \leq X} p^{-r}$$

whenever the sum diverges, while only finitely many such primes occur when $r \geq 2$. As a separately falsifiable genericity clause, for Kummer-generic Γ and $B = \{0\}$ one has $\kappa_{\Gamma,B} = 1$. The transition from Haar universality to the arithmetic intensity occurs only when w_p remains bounded.

Why this is top-tier. This separates two universality classes that a single undifferentiated model conflates. Growing targets are controlled by Haar geometry, while exact Wieferich targets can carry a genuinely arithmetic regulator intensity. Gras's much rarer fixed-base model and the Haar model therefore disagree precisely at the lattice boundary [10, 11]. The first theorem is mesoscopic universality for $w_p = p^\eta$ with one rank-one base.

For a p -adic unit a , write $\langle a \rangle_p$ for the Teichmüller lift of its residue, and for a prime $\mathfrak{p} \mid p$ of a number field write $\tau_{\mathfrak{p}}$ for the Teichmüller lift in the completion at $\mathfrak{p}$. Define $q_{p,k}(a) = p^{-1} \log(a \langle a \rangle_p^{-1}) \pmod{p^k}$. The matrix entries below are the $\mathfrak{p}$ -adic analogues of $q_{p,k}$.

Conjecture 14 (Horizontal p -adic regulator-matrix law). *Let $K/\mathbb{Q}$ be a fixed Galois number field and let $\Gamma \leq \mathcal{O}_K^\times$ be a saturated subgroup of rank s , generated by $\varepsilon_1, \dots, \varepsilon_s$. For a rational prime p splitting completely in K , order the primes $\mathfrak{p} \mid p$ by the Galois action and define the finite-log matrix to depth k by*

$$\mathcal{L}_{p,k}(\Gamma) = \left(p^{-1} \log_{\mathfrak{p}}(\varepsilon_j \tau_{\mathfrak{p}}(\bar{\varepsilon}_j)^{-1}) \pmod{p^k} \right)_{\mathfrak{p} \mid p, 1 \leq j \leq s}.$$

It lies in the exact linear relation module $\mathcal{R}_{\Gamma,k}$ imposed by the norm, the Galois action, and the integral multiplicative relations in Γ .

As p varies over rational primes splitting completely in K , the normalized matrices $\mathcal{L}_{p,1}/p$ equidistribute with Haar measure on the real relation torus attached to $\mathcal{R}_{\Gamma,1}$, and the depth- k lifts are Haar on $\mathcal{R}_{\Gamma,k}$. More generally, if $\mathcal{D}$ is a determinantal or Smith stratum whose reduction is transverse to the relation module and has codimension c , then the primes for which $\mathcal{L}_{p,k}$ lies in $\mathcal{D}$ obey the corresponding Haar local density. For codimension-one strata, the point process in $\log \log p$ converges after a uniform random translation to the Poisson law with the Haar intensity. Codimension at least two gives a finite exceptional set with the exact small-prime atoms predicted by the local measures.

Why this is top-tier. This turns mod- p Leopoldt phenomena into a horizontal random-matrix problem with the correct global relation module built in. Determinantal codimension, rather than the number of displayed units, controls rare regulator defects, while the Smith strata retain higher p -adic depth. Existing work supplies the regulator and random-matrix languages separately [11, 12, 13]. Their horizontal arithmetic identification is the new content. The first theorem is the rank-one real-quadratic split-prime law. The companion diagnostic verifies the norm relation exactly and finds fixed-frequency Fourier magnitudes at most 0.020 for $\mathbb{Q}(\sqrt{2})$, while making no claim about the matrix law.

Conjecture 15 (Functorial horizontal finite logarithms on algebraic tori). *Let $T/\mathbb{Q}$ be an algebraic torus, let $\mathcal{T}$ be its smooth integral model away from a finite set, and let $\Gamma \leq T(\mathbb{Q})$ be finitely generated. Let S be the identity component of the Zariski closure of Γ . At a good unramified prime p , let $\tau_p(\bar{P})$ be the canonical prime-to- p torsion lift of the reduction of $P \in \Gamma$ and define*

$$\ell_p(P) = p^{-1} \log_T(P \tau_p(\bar{P})^{-1}) \pmod{p} \in \text{Lie}(S)(\mathbb{F}_p).$$

For any generators $P_1, \dots, P_s$ of Γ , the normalized matrix $(\ell_p(P_i)/p)_i$ equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic

subgroup S and the integral relations among the P_i . For every morphism of tori $\phi : T \rightarrow T'$,

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.

Why this is top-tier. Katz already proposed Wieferich equidistribution beyond $\mathbb{G}_m$ [8]. The additional content is a matrix-valued law for a finitely generated global subgroup, relation-subtorus support, fixed-Frobenius disintegration, and exact functoriality under morphisms and isogenies. It turns horizontal finite logarithms into a functor on the category of tori with arithmetic subgroups.

6 Programme IV: arboreal arithmetic dynamics

For a degree- d map f and basepoint a , write $K_n = \mathbb{Q}(f^{-n}(a))$, $G_n = \text{Gal}(K_n/\mathbb{Q})$, and let $\mu_n(C) = |C|/|G_n|$ on conjugacy classes. For primes $p \leq X$ unramified in K_n , let $\hat{\mu}_{n,X}$ be the empirical Frobenius measure.

Conjecture 16 (Entropy-conductor universality of arboreal resolution). *Fix $0 < \eta < 1$. For a one-parameter family $\mathcal{F}$ of non-PCF (not postcritically finite) degree- d maps, let $G_{n,f}$ and $\mu_{n,f}$ be the level- n arboreal quotient and its Haar conjugacy-class measure for the parameter f . Sample parameters of height at most B and define*

$$R_{\mathcal{F}}(B, X; \eta) = \max \left\{ n : \mathbb{E}_{f \in \mathcal{F}(B)} d_{\text{TV}}(\hat{\mu}_{n,X,f}, \mu_{n,f}) \leq \eta \right\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, GRH holds for the preimage fields, and (B, X, n) lies in the family large-sieve range of Conjecture 17. Then

$$R_{\mathcal{F}}(B, X; \eta) = \log_d \log X + O_{\mathcal{F}, \eta}(1).$$

The $O(1)$ term is determined by the limiting Rényi-entropy profile of the nonuniform class measures and by the degree-normalized root-discriminant rate $d^{-n}[K_{n,f} : \mathbb{Q}]^{-1} \log |D_{K_{n,f}}|$, which stays bounded because $\log |D_{K_{n,f}}|/[K_{n,f} : \mathbb{Q}]$ grows like the critical-orbit height scale d^n , whereas $\log |D_{K_{n,f}}|$ itself is doubly exponential in n .

If two such families have the same degree, open generic images, uniformly comparable degree-normalized root-discriminant rates, and boundedly different class-measure Rényi profiles, then their observable depths differ by $O(1)$. In particular passage to a fixed-index open subgroup changes the resolution boundary by only a bounded number of levels.

Why this is top-tier. The conjecture identifies a phase invariant of an infinite Galois representation: how many levels can be reconstructed from primes below X . The leading $\log_d \log X$ scale comes from doubly exponential tree complexity, while the bounded shift records group index and ramification. Large arboreal images and effective Chebotarev are inputs, not the conclusion [14, 15, 16].

Conjecture 17 (Arboreal family large sieve at growing depth). *Let $\mathcal{F} \rightarrow \mathbb{A}^1$ be a generically finite-index arboreal family, and let $\mathfrak{q}_n$ be the conductor of the finite étale cover over the parameter line that records the level- n preimage tree. For a good prime p , a conjugacy class $C \subset G_n$, and a parameter c , put*

$$\Theta_{p,C}(c) = \mathbf{1}_{\text{Frob}_{p,c} \in C} - \mu_n(C).$$

For every $\varepsilon > 0$ and all complex coefficients $\alpha_{p,C}$,

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{C \subset G_n} \alpha_{p,C} \Theta_{p,C}(c) \right|^2 \ll_{\varepsilon, \mathcal{F}} (B + X^2 \mathfrak{q}_n^{1+\varepsilon}) \sum_{p,C} |\alpha_{p,C}|^2 \mu_n(C),$$

uniformly for n with $X^2 \mathfrak{q}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$. The same bound holds for bounded cylinder functions involving every level $j \leq n$.

In the critical range where the analytic Chebotarev error is $o(1)$ but the iid sampling functional of μ_n is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from μ_n , including its missing-mass and collision profile.

Why this is top-tier. This is the quantitative arithmetic engine missing from purely group-theoretic resolution heuristics. It extends the Frobenius large-sieve philosophy of Kowalski to growing arboreal quotients and simultaneously controls parameter and prime dependence [17]. A counterexample would exhibit systematic cross-prime or cross-parameter entanglement invisible in the generic arboreal group.

Conjecture 18 (Coloured dynatomic Galois-factor process). *For $f_c(x) = x^d + c$, let the critical exact-period polynomial factor as $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$ over $\mathbb{Z}$. Sample c from a dyadic height interval away from PCF, discriminant, and fixed-resultant loci.*

For every fixed n , condition on any fixed finite local valuation state and normalize the prime factors of each $F_{n,j}(c)$ by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet PD(1) processes, with the mark in colour j given by the fixed-point-size-biased conjugacy law in the splitting field of $F_{n,j}$. The total logarithmic mass of ramified factors tends to zero.

Uniformly for $n = n(H)$ with $\deg \Psi_n = o(\log H)$, every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.

Why this is top-tier. Exact-period components are genuine geometric colours, and their Galois marks distinguish dynamical families with similar unmarked factor sizes. The statement deliberately separates the full fixed-degree process from the restricted-support growing-degree law, in accordance with the distribution thresholds in Bharadwaj–Rodgers [21]. The first theorem is a two-colour correlation law at one fixed period.

Conjecture 19 (Primitive valuation process for dynatomic values). *For every level n , first remove from the exact-period polynomial Ψ_n the canonical greatest common divisor in $\mathbb{Q}[c]$ with the earlier critical-orbit product $\prod_{m < n} f_c^m(0)$, and call the resulting polynomial Ψ_n^{new} . For each prime p , let $\nu_{n,p}$ be the exact Haar law on $\mathbb{Z}_p$ of the primitive valuation*

$$V_{n,p}(c) = v_p(\Psi_n^{\text{new}}(c)) \mathbf{1}_{p \nmid f_c^m(0) \ \forall m < n}.$$

All discriminant, resultant, and collision primes are carried explicitly by these exact local laws.

If $n = n(H)$ satisfies $\deg \Psi_n = o(\log H)$ and the local squarefree product is nonzero, then the full vector $(V_{n,p}(c))_p$, for c in a dyadic height interval, converges uniformly in n to the adelic product $\bigotimes_p \nu_{n,p}$ in the topology generated by bounded cylinder functions and the squarefull-mass functional

$$\sum_p (\log p) \max(V_{n,p} - 1, 0).$$

Consequently the new primitive part is squarefree with probability $\prod_p \nu_{n,p}(\{0,1\}) + o(1)$, and its logarithmic squarefull mass converges to the compound-Bernoulli law determined by the complete local measures. The same process law is uniform under fixed algebraic specializations that preserve the local product.

Why this is top-tier. This predicts the complete primitive valuation process, not merely one squarefree probability. The level dependence is essential: for fixed p , primitive hits occur by a bounded time modulo p^k . Exact bad-prime and collision behaviour is retained in the local measures, while only genuine polynomial common factors are removed canonically. The process supplies the multiplicity input needed for dynatomic factor laws. The nearest literature treats primitive divisors and fixed-polynomial squarefree phenomena, not this uniform adelic process [19, 18].

Conjecture 20 (Divisor-sensitive classification of dynamical gcd height). *Let $F = (f, g)$ be a split map on $(\mathbb{P}^1)^2$, with f, g disintegrated (in the sense of Medvedev and Scanlon [26], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let $P = (a, b)$ have positive canonical heights. Let D_1, D_2 be the two coordinate-zero divisors and let E be the exceptional divisor of the blow-up at $D_1 \cap D_2$. Then*

$$\limsup_{n \rightarrow \infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

if and only if there exist $m \geq 0$ and an irreducible F -periodic curve Z containing $F^m(P)$ such that, on the normalization of Z , the pullbacks of D_1 and D_2 have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.

The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on $(\mathbb{P}^1)^r$ by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.

Why this is top-tier. The obstruction is not merely a common dynamical quotient. It is a periodic geometric relation carrying the actual divisors measured by the gcd height. Eventual entry is the right notion, since a preperiodic tail can carry the divisor relation before periodicity begins. Recent work resolves several set-theoretic dynamical-GCD problems, while this height-theoretic divisor classification is the additional boundary [20]. Failure would point to a new source of macroscopic Diophantine proximity outside invariant geometry.

7 Programme V: adelic factorization and sieve flow

Throughout this programme $P^+(m)$ and $P^-(m)$ denote the largest and smallest prime factors of m , the constant γ is Euler's, $\rho_f(p)$ counts the roots of f modulo p , and every sample $n \in [N, 2N]$ is drawn uniformly.

Conjecture 21 (Frobenius-marked Poisson–Dirichlet process). *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and transitive Galois group G . Sample $n \in [N, 2N]$. Mark each unramified prime factor p of $f(n)$ by its conjugacy class $C \subset G$ and give it mass $u = \log p / \log |f(n)|$. The macroscopic marked process converges to a marked PD(1) process with conditional mark law*

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|} \text{fix}(C),$$

independent of u . Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.

Why this is top-tier. Chebotarev contributes $|C|/|G|$, divisibility by a polynomial value size-biases by the number of fixed roots, and Burnside's lemma normalizes the result. Bharadwaj–Rodgers supply the unmarked factor-process framework [21]. Simultaneous Galois marking, conditional independence, and functorial pushforward are the new content. This is the cleanest bridge in the collection between Galois theory and probabilistic factorization.

Conjecture 22 (Three-scale factorization of polynomial values). *Let $f \in \mathbb{Z}[x]$ be irreducible, sample $n \in [N, 2N]$, choose $y = y(N) \rightarrow \infty$ with $\log y = o(\log N)$, and write $R_y = f(n)/S_y(f(n))$, where $S_y(m) = \prod_{p \leq y} p^{v_p(m)}$ is the y -smooth part. Jointly with the complete small-prime valuation field $(v_p(f(n)))_{p \leq y}$, the residual logarithmic factor process*

$$\sum_{p|R_y} v_p(R_y) \delta_{(\log p / \log |R_y|, \text{Frob}_p)}$$

converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law, and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity du/u , marked by Conjecture 21, conditioned to have total mass one. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.

For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.

Why this is top-tier. Raw small–large independence is false because small factors consume mass. This conjecture identifies the complete conservation-corrected product structure and adds the mesoscopic scale that is absent from a two-scale statement. It would justify, in one limit theorem, the local sieve model, the scale-invariant factor process, and the macroscopic partition.

Conjecture 23 (Adelic Gibbs gluing for reducible polynomial values). *Let $f = \prod_{i=1}^s g_i$ be squarefree with irreducible components, sample $n \in [N, 2N]$, and put $t = n/N$. For each prime p , let ν_p be the exact Haar law on $\mathbb{Z}_p$ of the valuation vector $(v_p(g_1(x)), \dots, v_p(g_s(x)))$. For $y \rightarrow \infty$ with $\log y = o(\log N)$, consider stable convergence jointly with the archimedean coordinate t and condition on $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$. After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 22, with the marks of Conjecture 21.*

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in t and the adelic local law $\bigotimes_p \nu_p$, in the iterated order $N \rightarrow \infty$ and then $y \rightarrow \infty$. Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx$$

together with the archimedean size profile $(\log |g_i(Nt)|)_i$. No additional colour coupling survives after this adelic–archimedean mixture.

Why this is top-tier. A scalar Euler factor cannot specify a point process. This statement gives the full local-to-global gluing rule and includes the archimedean variable that can couple component sizes. It is functorial under multiplying, splitting, or regrouping components. A persistent medium-prime or archimedean coupling after the stated mixture would identify a new nonadelic invariant.

Conjecture 24 (Full multivariate adelic saddle law for joint smoothness). *Let $f_1, \dots, f_s \in \mathbb{Z}[x]$ be pairwise coprime and primitive, sample $n \in [N, 2N]$, and put $t = n/N$. For complex variables $\mathbf{s}$ and smoothness bounds $\mathbf{y}$, define*

$$\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i:p \leq y_i} (s_i-1)v_p(f_i(x))} dx, \quad \widehat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) = \frac{\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s})}{\mathcal{Z}_{p,\mathbf{y}}(+\infty)},$$

where $\mathcal{Z}_{p,\mathbf{y}}(+\infty)$, the limit as every $s_i \rightarrow +\infty$, is the Haar probability that p divides no $f_i(x)$ with $p \leq y_i$, and the exact multivariate saddle action is

$$\Phi_t(\mathbf{s}) = \sum_p \log \widehat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

The normalization is fixed by the one-variable calibration: for $f(x) = x$ the local factors $\widehat{\mathcal{Z}}_{p,y}(s) = (1 - p^{-s})^{-1}$ and the action reduce exactly to the Hildebrand–Tenenbaum saddle for the smooth-number density [25], whereas the unnormalized probability partition function would insert the spurious sieve factor $\prod_{p \leq y} (1 - 1/p)$, and the omitted Perron kernel $\prod_i s_i^{-1}$ would return the cumulative average rather than the local density. Let $\boldsymbol{\sigma}_t$ be the unique real saddle in the admissible region. Define the joint saddle density by the independently specified inverse-Mellin integral

$$\mathcal{D}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\boldsymbol{\sigma}_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} d\mathbf{s},$$

with the standard truncated-contour limit, and let $H_t = \nabla^2 \Phi_t(\boldsymbol{\sigma}_t)$. Uniformly when every $u_i(t) = \log |f_i(Nt)| / \log y_i$ remains in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \leq y_i \ \forall i) \sim \int_1^2 \mathcal{D}_t^{\text{joint}} dt,$$

and the integral admits the full Gaussian saddle expansion with action $\Phi_t(\boldsymbol{\sigma}_t)$ and curvature matrix H_t , with $\mathcal{D}_t^{\text{joint}}$ the local smooth-value density at t .

If $\mathcal{D}_{i,t}^{\text{marg}}$ are the one-polynomial saddle densities, then $\mathcal{D}_t^{\text{joint}} / \prod_i \mathcal{D}_{i,t}^{\text{marg}}$ is asymptotic to the complete ratio of joint and marginal actions and Hessian determinants. Its Euler-product part is the ratio of the exact p -adic partition functions evaluated at the joint and marginal saddles. This local factor, the saddle displacement, and the mixed curvature are all indispensable at leading order.

Why this is top-tier. This repairs the principal weakness of simpler joint-smoothness heuristics: the joint saddle and its curvature cannot be replaced by a scalar Euler correction evaluated at marginal saddles. The conjecture combines exact higher p -adic valuations, moving archimedean sizes, and multivariate saddle geometry. Even the one-polynomial theory is difficult [22, 23], and the full joint adelic action would be a major new local–global bridge.

Conjecture 25 (Buchstab–Bateman–Horn component flow). *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible, and admissible, and sample $n \in [N, 2N]$. For y with $u_n = \log |f(n)| / \log y$ in a fixed compact subset of $(1, \infty)$, let $\Omega_y(f(n))$ be the number of prime factors of $f(n)$ exceeding y , counted with multiplicity, on the event $P^-(f(n)) > y$. Put*

$$V_f(y) = \prod_{p \leq y} \left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{\substack{t_1, \dots, t_{j-1} \geq 1 \\ t_1 + \dots + t_{j-1} \leq u-1}} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

so that $\sum_{j \geq 1} \omega_j(u) = \omega(u)$. Then for every fixed j ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact u -ranges. Conditional on the ordered logarithmic sizes of the j remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 21. Summing over j gives the rough-value Buchstab law, $j = 1$ gives the Bateman–Horn prime density, and $j = 2$ gives the semiprime transition. Whenever $\max_{N \leq n \leq 2N} |f(n)| < y^2$, every rough value has exactly one prime factor above y , so the $j = 1$ component exhausts the rough set, while the density formula itself remains conjectural.

Why this is top-tier. The conjecture resolves a rough polynomial value into its complete finite- u factor count and Galois mark content. Primes and semiprimes are not inserted as separate heuristics. They are components of one universal flow. Generalized Buchstab equations are the random-integer antecedent [27], and the fixed-polynomial, locally corrected, Galois-marked component law is the new statement. It is strongly falsifiable even when total roughness follows Buchstab.

8 Adversarial quality audit

The set is intentionally not a list of equally likely claims, but every numbered entry passes the same top-tier gate. Every numbered statement except Conjectures 2 and 5 is a parent mechanism, while those two are classification laws with independent content rather than generic consequences.

The first decisive theorem for each programme is deliberately much smaller than the full conjecture:

1. Compute the complete connected third cumulant for two pair motifs.
2. Prove the two-point connected arrival kernel K_2 and its additive polyspectrum.
3. Prove a rank-one positive-power finite-logarithm large sieve.
4. Prove the arboreal family large sieve at one growing quadratic level.
5. Prove the two-component case of Conjecture 23 and the $j = 2$ Buchstab component in an averaged polynomial family.

Novelty caution. The literature screening establishes nearest visible boundaries, not priority. In particular, Katz already proposed broad algebraic-group Wieferich equidistribution, Bharadwaj–Rodgers already provide the unmarked large-factor process under distribution hypotheses, Wood already treats fixed local class-group conditions, and recent work on dynamical gcds is rapidly developing. The claimed additional content is therefore the specific relation-torus, marked, process-level, growing-condition, or classification structure stated above.

References

- [1] H. Montgomery and K. Soundararajan, *Primes in short intervals*, Comm. Math. Phys. 252 (2004), arXiv:math/0409258.
- [2] T. H. Chan, *More precise pair correlation of zeros and primes in short intervals*, arXiv:math/0206292.
- [3] V. Kuperberg, *Odd moments in the distribution of primes*, arXiv:2109.03767.
- [4] V. Kuperberg, B. Rodgers and E. Roditty-Gershon, *Sums of singular series and primes in short intervals in algebraic number fields*, arXiv:2001.09513.
- [5] J. Li, K. Pratt and G. Shakan, *A lower bound for the least prime in an arithmetic progression*, arXiv:1607.02543.
- [6] M. Rubinstein and P. Sarnak, *Chebyshev's bias*, Experimental Mathematics 3 (1994), 173–197.
- [7] M. M. Wood, *Cohen–Lenstra heuristics and local conditions*, arXiv:1710.01350.
- [8] N. M. Katz, *Wieferich past and future*, Contemporary Mathematics 632 (2015), 253–270.
- [9] I. E. Shparlinski, *Fermat quotients: exponential sums, value set and primitive roots*, arXiv:1104.3909.
- [10] G. Gras, *Étude probabiliste des quotients de Fermat*, Funct. Approx. Comment. Math. 54 (2016), arXiv:1409.2815.
- [11] G. Boeckle, D.-A. Guiraud, S. Kalyanswamy and C. Khare, *Wieferich primes and a mod p Leopoldt conjecture*, arXiv:1805.00131.
- [12] Y. Wang and R. P. Stanley, *The Smith normal form distribution of a random integer matrix*, arXiv:1506.00160.
- [13] R. Van Peski, *Limits and fluctuations of p -adic random matrix products*, arXiv:2011.09356.
- [14] R. Jones, *Galois representations from pre-image trees: an arboreal survey*, arXiv:1402.6018.
- [15] B. Kadets, *Large arboreal Galois representations*, arXiv:1802.09074.
- [16] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, arXiv:1803.02823.
- [17] E. Kowalski, *The large sieve, monodromy and zeta functions of curves*, J. reine angew. Math. 601 (2006), 29–69, arXiv:math/0503714.
- [18] D. Krumm, *Galois groups in a family of dynatomic polynomials*, arXiv:1707.02501.
- [19] K. Doerksen and A. Haensch, *Primitive prime divisors in zero orbits of polynomials*, arXiv:1009.3971.
- [20] S. Yang and X. Zhong, *Dynamical GCD problems and a variant of the Dynamical Mordell–Lang conjecture*, arXiv:2602.18302.
- [21] A. Bharadwaj and B. Rodgers, *Large prime factors of well-distributed sequences*, arXiv:2402.11884.
- [22] G. Martin, *An asymptotic formula for the number of smooth values of a polynomial*, arXiv:math/9909180.

- [23] M. Mine, *An upper bound for the number of smooth values of a polynomial and its applications*, arXiv:2410.09558.
- [24] D. A. Goldston and H. L. Montgomery, *Pair correlation of zeros and primes in short intervals*, in Analytic Number Theory and Diophantine Problems, Progr. Math. 70, Birkhäuser, 1987, 183–203.
- [25] A. Hildebrand and G. Tenenbaum, *On integers free of large prime factors*, Trans. Amer. Math. Soc. 296 (1986), 265–290.
- [26] A. Medvedev and T. Scanlon, *Invariant varieties for polynomial dynamical systems*, Ann. of Math. 179 (2014), 81–177, arXiv:0901.2352.
- [27] P. Moree, *A generalization of the Buchstab equation*, Manuscripta Math. 94 (1997), 267–270.